

Coherent Free-Space Optical Communications in Atmospheric Turbulence: Principles, Limits, and Emerging Strategies

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Abstract *Coherent free-space optical communication is fundamentally limited by atmospheric turbulence. This tutorial presents a unified statistical modal framework for analyzing performance limits and reviewing mitigation strategies, highlighting statistically matched transmission modes that enable robust and scalable link design without instantaneous wavefront correction. ©2026 The Author*

Introduction and Motivation

Coherent free-space optical (FSO) communication systems offer high sensitivity and spectral efficiency, but are fundamentally limited by atmospheric turbulence, which induces random amplitude fluctuations and phase distortions that degrade signal coherence. Atmospheric turbulence originates from random refractive index variations along the propagation path, leading to beam spreading, scintillation, and phase aberrations [1].

In coherent systems, these effects are particularly detrimental, as both amplitude and phase fluctuations directly impact detection performance [2]. The resulting channel is spatially coupled and time-varying, making deterministic compensation increasingly challenging in practical systems, particularly under strong or rapidly varying turbulence conditions. Conventional mitigation strategies have traditionally relied on real-time wavefront correction, requiring complex hardware and instantaneous channel estimation. In contrast, this tutorial adopts an alternative approach that exploits the statistical structure of the turbulent optical field to design transmission and reception strategies that are inherently robust to channel fluctuations.

From this physical viewpoint, it is useful to adopt a system-level interpretation of how turbulence affects coherent propagation. Turbulence can be interpreted as inducing coupling among orthogonal spatial modes of the optical field defined over the receiver aperture [3]. This perspective offers an intuitive picture of turbulence-induced coupling, as it redistributes optical energy among spatial degrees of freedom rather than simply attenuating the signal.

Building on this interpretation, the impact of turbulence on system performance can be quantified in terms of key communication metrics. The performance of coherent FSO links can be understood in terms of power efficiency, link reliability, and bandwidth efficiency [4]. Power efficiency is determined by the achievable signal-to-noise

ratio, link reliability is governed by fading statistics, and bandwidth efficiency is affected by modal crosstalk in multiplexed systems. Because these metrics are inherently coupled under turbulence, this tutorial adopts a statistical modal perspective to analyze the resulting trade-offs and to identify robust transmission and reception strategies.

Turbulence Effects and Fundamental Limits

Atmospheric turbulence induces random transformations of the optical field that can be modeled as a coupling process among spatial modes. In this framework, turbulence is modeled as a random coupling process among spatial modes, such that the received field becomes a superposition of orthogonal modes with random coefficients determined by the channel statistics. This modal description reveals that turbulence does not only attenuate the signal, but spreads energy across modes according to its spatial coherence properties and propagation geometry.

A key element in this description is the spatial correlation function of the optical field over the receiver aperture, which captures the second-order statistics imposed by turbulence [5]. This correlation structure determines how efficiently different spatial components of the transmitted field are preserved or mixed during propagation. As turbulence strength increases, the effective number of significant modes grows, reflecting a loss of spatial coherence and a corresponding increase in channel dimensionality.

This viewpoint leads directly to fundamental limits on system performance. Power efficiency is maximized when the transmitted field aligns with the dominant mode of the channel, which captures the largest fraction of the coherent energy transfer and thus maximizes the average received signal-to-noise ratio.

However, such concentration of power is inherently sensitive to fading, since fluctuations in that dominant mode directly translate into fluctuations in the received signal. Link reliability,

in contrast, is improved by distributing power across multiple modes, providing diversity against turbulence-induced fluctuations at the cost of reduced average signal strength per mode. In spatial-mode multiplexing systems [6][7], turbulence further degrades bandwidth efficiency by inducing inter-modal crosstalk, reducing the effective number of orthogonal channels and increasing detection complexity.

These observations highlight a fundamental trade-off between efficiency and robustness. Strategies that maximize average performance tend to be sensitive to fluctuations, while those that improve reliability sacrifice peak efficiency. This trade-off is a direct consequence of the statistical redistribution of energy across modes imposed by turbulence and cannot be eliminated, only managed — a design challenge that motivates the strategies developed throughout this tutorial.

Mitigation Strategies: From Instantaneous Correction to Statistical Design

A broad range of mitigation techniques has been developed to address turbulence-induced impairments. Adaptive optics represents the most direct approach, aiming to correct phase distortions in real time through wavefront sensing and compensation [8][9]. From a conceptual standpoint, adaptive optics attempts to invert the turbulence-induced transformation of the optical wavefront, restoring a near-ideal coherent field at the receiver. When effective, this approach can significantly improve signal quality and recover much of the original beam coherence. However, it relies on accurate and fast channel estimation and becomes increasingly complex as system size and turbulence strength grow. Moreover, it primarily addresses phase distortions and may be limited by amplitude fluctuations and temporal decorrelation of the channel.

An alternative approach avoids full wavefront reconstruction by exploiting spatial diversity [10][11]. In coherent array receivers, the optical field is collected over multiple subapertures, each coupled to a detector or single-mode fiber. In contrast to adaptive optics, these systems do not attempt to invert the turbulence transformation, but instead sample it across multiple spatial realizations. By adjusting amplitude and phase weights across array elements, they coherently combine these samples to average the effects of mode coupling rather than eliminate it, partially compensating for turbulence-induced fading and improving effective coupling efficiency. While this approach improves robustness under stronger turbulence conditions, it introduces complexity in calibration, synchronization, and signal combining as the number of apertures increases.

Modal compensation strategies provide an intermediate perspective by representing the received field in an orthogonal basis and selectively processing dominant components [12][13]. In this framework, turbulence-induced distortions are represented in a modal basis as a redistribution of energy among spatial modes. Rather than inverting the full transformation, modal techniques attempt to reconstruct the field within a reduced subspace spanned by the most significant modes [6]. This enables partial recovery of the coherent signal and offers a flexible trade-off between performance and complexity. However, like adaptive optics, these methods typically rely on instantaneous or adaptive channel information to identify and compensate the relevant modal components, which limits their scalability in dynamic environments.

Despite their differences, all three approaches rely — explicitly or implicitly — on instantaneous channel knowledge or partial field reconstruction, which introduces fundamental limitations in complexity, scalability, and robustness under strong or rapidly varying turbulence.

This observation motivates a complementary design philosophy in which system optimization is not based on instantaneous channel inversion, but on the statistical structure of the optical field. In this framework, the channel is not treated as a deterministic object to be corrected, but as a stochastic operator whose second-order properties — in particular, the spatial correlation structure of the turbulent field — can be exploited to define optimal transmission and reception bases.

Statistically Matched Modal Design

A fundamentally different design paradigm emerges when system optimization is based on the statistical properties of turbulence [14] rather than its instantaneous realization. In this framework, the optical field is characterized through its spatial correlation function, which defines an eigenvalue problem whose solutions correspond to orthogonal modes that diagonalize the channel statistics. These modes form the Karhunen–Loève (KL) expansion [15] of the optical field and provide an optimal basis for minimum mean-square error representation.

The KL framework reveals that only a limited number of modes carry most of the transmitted energy after propagation, enabling a reduction of the effective channel dimensionality and providing a principled basis for designing transmission and reception strategies that focus on the most significant modes. In near-field propagation, these optimal modes can be closely approximated by Laguerre–Gaussian modes with

an adaptive beam waist determined by the coherence properties of the field at the receiver plane [3]. Such adaptive-waist modes provide a practical means to approach statistical optimality while remaining compatible with physically realizable optical systems.

Within this statistical description, the trade-off between power efficiency and link reliability can be understood in a unified manner [4]. Concentrating power in the dominant mode maximizes the average received signal-to-noise ratio, as it aligns the transmitted field with the most stable component of the channel. Distributing power across multiple dominant modes, on the other hand, introduces diversity that reduces sensitivity to turbulence-induced fading and improves link robustness. The optimal balance between these strategies depends on system requirements and operating conditions, but the modal framework provides a clear and physically grounded way to navigate this trade-off.

A key advantage of statistically matched modal design is that it does not require instantaneous channel knowledge. Instead, it relies on slowly varying or known turbulence statistics, which can be estimated over longer time scales or derived from propagation models. This significantly reduces system complexity while maintaining high performance, making the approach particularly attractive for the large-scale or rapidly varying systems where, as discussed above, real-time wavefront correction becomes impractical.

Conclusions

We have presented a unified perspective on coherent free-space optical communication under atmospheric turbulence, based on a statistical modal framework. Turbulence

fundamentally alters system performance through structured mode coupling, giving rise to inherent trade-offs between power efficiency and link reliability.

Conventional mitigation strategies can be interpreted as different ways of addressing the turbulence-induced modal coupling operator. Adaptive optics attempts to invert this operator through instantaneous wavefront correction, restoring the coherent field when accurate channel estimation is available. Array-based receivers, in contrast, sample the distorted field across multiple spatial realizations and mitigate its impact through coherent combining, effectively averaging the effects of the coupling. Modal compensation strategies operate in a reduced basis, capturing and reconstructing the dominant components of the coupled field within a reduced basis, thereby truncating the effective channel representation.

In contrast to these instantaneous approaches, the statistical design paradigm presented in this tutorial unifies a wide range of existing techniques and provides a principled foundation for the development of robust and scalable coherent FSO systems, in which the channel is characterized through its second-order properties rather than its instantaneous realization. In this context, Karhunen–Loève modes define optimal transmission and reception bases that diagonalize the statistical structure of turbulence. Within this framework, concentrating power in dominant modes maximizes efficiency, while distributing power across multiple modes enhances reliability.

Ultimately, the path to robust coherent FSO systems lies not in correcting the turbulent channel, but in aligning transmission and reception with its underlying statistical structure.

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